



Detecting Careless Respondents in Multidimensional Forced Choice Data: An Application of I_z Person-Fit Statistic to the TIRT Model

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Abstract

There has been growing interest in the use of multidimensional forced-choice (MFC) measures in industrial and organizational psychology. However, despite the growing popularity, there has been a lack of research on how to address the problem of careless responses and improve the quality of MFC data. This study aims to address this gap by discussing the issue of careless responses in MFC data, proposing a method for detecting careless respondents using the I_z person-fit statistic in conjunction with the Thurstonian item response theory (TIRT; Brown & Maydeu-Olivares, 2011) model, and examining the performance of this method on pair and triplet MFC measures and its impact on psychometric properties. To this end, we conducted empirical and simulation studies under various experimental conditions. We also provided an R tutorial for applying the I_z person-fit statistic to both pair and triplet MFC measures using the TIRT model. The results showed that our proposed method can effectively detect careless respondents without a significant risk of misclassification. The implications of the findings and practical guidelines were discussed.

Keywords Multidimensional forced-choice measure · Thurstonian item response theory · Personality · Careless response · Self-report measure · I_z person-fit statistic

Over the past two decades, there has been increasing interest in the use of multidimensional forced-choice (MFC) measures in industrial and organizational (I-O) psychology (Cao & Drasgow, 2019; Wetzel et al., 2020). MFC measures typically consist of two (i.e., pair), three (i.e., triplet), or four (i.e., tetrad) statements within an item block, each of the statements measures a distinct psychological construct. Statements within an item block are matched based on social desirability. Respondents are then forced to rank the statements in each block from “most like me” to “least like me”. An example of an MFC triplet item is shown below:

Rank the three statements in each group from “most like me (1)” to “least like me (3)”.

- A. *I do things according to a plan.* 1
- B. *I make friends easily.* 2
- C. *I enjoy hearing new ideas.* 3

In this example, the three statements within the item block measure Conscientiousness, Extraversion, and Openness to Experience and were ranked 1st, 2nd, and 3rd, respectively.

When presented with such a choice, respondents may not easily fake their responses if MFC measures are properly designed, because all statements presented within an item block are equally socially desirable (or undesirable). With this advantage, MFC measures have gained increasing attention in applied and academic settings, including the Occupational Personality Questionnaire (Brown & Bartram, 2009), the Personality and Preference Inventory (Talogy, 2023), the Global Personality Inventory–Adaptive (CEB, 2010), the Tailored Adaptive Personality Assessment System (Stark et al., 2014), the Emotional Intelligence Questionnaire (Anguiano-Carrasco et al., 2015), the Adaptive Employee Personality Test (Boyce et al., 2015), the Big

Additional supplementary materials may be found here by searching on article title <https://osf.io/collections/jbp/discover>

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Five Personality MFC measure (Lee et al., 2018; Wetzel & Frick., 2020), the Maladaptive Personality Assessment (Guenole et al., 2018), and the CIVIC MFC Assessment (Ng et al., 2021).

To date, much of the research on MFC measures has largely focused on five main areas: (i) faking, (ii) validity, (iii) test-taker reaction, (iv) item response theory (IRT) model development, and (v) item or key matching. Specifically, research has shown that MFC measures can effectively reduce faking responses (e.g., Cao & Drasgow, 2019; Christiansen et al., 2005; Fisher et al., 2018; Hirsh & Peterson, 2008; Jackson et al., 2000; Lee & Joo, 2021; Lee et al., 2019a, b; Martin et al., 2002; Martínez and Salgado, 2021; O'Neill et al., 2017; Vasilopoulos et al., 2006; Wetzel et al., 2021), provide similar or better criterion-related validity compared to Likert-type measures (e.g., Bartram, 2007; Chernyshenko et al., 2009; Converse et al., 2008; Lee et al., 2018; Otero et al., 2020; Salgado & Tauriz, 2014; Salgado et al., 2015; Wetzel & Frick, 2020; Zhang et al., 2020), result in more negative test-taker reactions than the Likert-type format (e.g., more cognitively demanding, fatiguing, and uninteresting; Bowen et al., 2002; Converse et al., 2008; Dalal et al., 2021; Sass et al., 2020), and can be estimated using various IRT models (e.g., Brown & Maydeu-Olivares, 2011; de la Torre et al., 2012; Joo et al., 2018, 2020; Lee et al., 2019a, 2019b; Morillo et al., 2016; Qiu & Wang, 2021; Stark et al., 2005; Wang et al., 2017). In addition, recent research has suggested that item desirability matching is a critical factor in designing MFC measures that are resistant to faking (Hughes et al., 2021; Kreitchmann et al., 2023; Lee et al., 2022; Lin et al., 2023; Pavlov et al., 2021). As such, there has been a substantial increase in research on various topics of MFC measures, but the issue of careless responses in MFC data has received little attention.

The problem of careless responses in survey data has received considerable attention in recent years (e.g., Arthur et al., 2021; Bowling & Huang, 2018; Breitsohl and Steidelmüller, 2018; Curran, 2016; Gibson & Bowling, 2019; Goldammer et al., 2020; Huang et al., 2015; Kam, 2019; Kung et al., 2018; Lovett et al., 2018; Meade & Craig, 2012; Ward & Meade, 2018, 2023). However, most studies on this topic have focused exclusively on traditional Likert-type measures. As emphasized by Bowling and Huang (2018), “more attention should be given to the effective measurement of careless responding” (p. 229) as a future research direction. In addition, Frick (2022) recently suggested that “future research could develop models for other response biases, such as *careless responding in MFC data*” (p.791). To date, various strategies have been proposed to identify careless respondents, such as longstring, even–odd consistency, and Mahalanobis distance (see Arthur et al., 2021; Meade & Craig, 2012; Ward & Meade, 2023). Each method is designed to identify a different type of careless

responses. For example, the longstring index is particularly well suited for detecting straight lining careless responses, the even–odd index is adept at identifying inconsistent or random responses, and the Mahalanobis method focuses on capturing abnormal patterns in the data (e.g., outliers). However, they are primarily designed for traditional Likert-type measures and cannot be directly applied to MFC measures because of their different format (i.e., forced choice) and response assumption (i.e., comparative judgment assumption). Therefore, it is imperative to develop a novel approach to detect careless respondents in MFC data.

Our research aims to fill this research gap by (a) discussing the issue of careless responding in MFC personality measures, (b) proposing the l_z person-fit statistic (Drasgow et al., 1985) in conjunction with the Thurstonian item response theory (TIRT; Brown & Maydeu-Olivares, 2011) to detect careless respondents, (c) examining the performance of the l_z person-fit statistic in terms of power and Type I error, (d) investigating the reliability and criterion-related validity improvements after removing identified careless respondents, and (e) providing an R tutorial and source code for the practical application of our method in the Appendix. To this end, we conducted both empirical and simulation studies.

This paper is organized as follows: First, we discuss the issue of careless responding in MFC measures. Then, we introduce the TIRT model and the application of l_z person-fit statistics. Next, we present our empirical and simulation study results and discuss their implications for applied and research settings.

Careless Response Issues of MFC Measures

Careless responding is a response set in which participants respond without much consideration, motivation, or pattern (Meade & Craig, 2012; Ward & Meade, 2023; Ward et al., 2017). The prevalence of careless respondents in survey data can vary, but the literature generally supports a range of prevalence from 10 to 15% (Curran, 2016), 33% (Goldammer et al., 2020), 46% (Oppenheimer et al., 2009), and even up to 60% (Hauser & Schwarz, 2016; Meade & Craig, 2012). Moreover, careless responses can negatively affect research findings by decreasing reliability (Arias et al., 2020), threatening construct validity (Kam, 2019), distorting factor analytic results (Huang et al., 2015), and attenuating statistical inferences (Maniaci & Rogge, 2014). These effects can ultimately compromise the validity of the research findings and lead to ineffective organizational decision-makings and management strategies (Goldammer et al., 2020; Schmitt et al., 1999). This suggests that MFC response data may also be influenced by careless responses, and underscores the importance of considering this issue when analyzing and interpreting research findings derived from MFC data.

Careless responses can occur in MFC data for various reasons. Ward and Meade (2023) recently identified three potential factors that can contribute to the presence of careless responses in survey data: i) the survey design, ii) the characteristics of the sample, and iii) the context in which the data is collected. In this paper, we discuss the potential sources of careless responses that may arise in MFC measures, based on these three factors. By understanding the potential sources of careless responses, researchers can take steps to identify and mitigate their effects, ensuring the accuracy, reliability, and validity of their findings.

Characteristics of the Survey (test) Design¹

The presence of careless responses in survey data can be influenced by various characteristics of the survey itself, such as its length, instructions, or test format (Meade & Craig, 2012; Ward & Meade, 2023). In contrast to Likert-type measures, MFC measures present multiple statements (e.g., two, three, or four) within an item block and require respondents to choose the statements that are most and least like them or rank the statements. In the triplet MFC format, respondents are expected to compare the first and second statements, the first and third statements, and the second and third statements, respectively (Brown & Maydeu-Olivares, 2011). This unique nature of the MFC format, in which a single triplet item involves three sets of comparative evaluations, increases cognitive load and test fatigue (Lee et al., 2021), potentially leading to careless responses.

In addition, MFC measures are typically recommended to include heteropolar blocks (i.e., the mixing of positively and negatively keyed statements within an item block) to improve the scoring accuracy (Brown & Maydeu-Olivares, 2011; Lee et al., 2022). The following example includes both positively and negatively keyed statements.

- A. *I do things according to a plan.*
- B. *I often feel uncomfortable around others.*
- C. *I enjoy hearing new ideas.*

The first and third statements are positively keyed statements that measure Conscientiousness and Openness, respectively. In contrast, the second statement is a negatively keyed statement that measures Extraversion. The inclusion of many heteropolar blocks may make MFC measures more cognitively demanding. As noted by Lee et al. (2022), heteropolar blocks “could yield a disruption in the item-response process because respondents have to frequently shift their

cognitive processing of the content depending on the keyed directions (*p. 3*)”. Consequently, the unique test design regarding heteropolar blocks could make comparative judgment more complicated, which may yield careless responses.

Ward and Meade (2023) also suggested that mandatory survey participation may cause respondents to lose interest in the survey, leading to careless responses. This is particularly problematic for MFC measures because respondents are *forced* to select or rank statements, even when respondents have no preferences within an MFC item. The forced nature of MFC measures may also contribute to the presence of careless responses by decreasing respondents' interest in the survey. Finally, the likelihood of careless responses may increase as the size of the MFC item block (e.g., pair, triplet, tetrad) and the length of the test increase due to respondents' fatigue (e.g., Galesic & Bosnjak, 2009; Gibson and Bowling, 2019; Lee et al., 2021).

Characteristics of the Sample

The characteristics of the sample can also contribute to the presence of careless responses in survey data (Meade & Craig, 2012; Ward & Meade, 2023). In particular, responses from students and online samples are more likely to be careless (e.g., Bowling & Huang, 2018; Jones et al., 2022; Lovett et al., 2018; Meade & Craig, 2012; Niessen et al., 2016). For example, students who participate in a study for course credit may lack interest in the research or a sense of responsibility to respond carefully, and online samples may be easily distracted. Even responses from crowdsourced samples from online survey panels can be prone to careless responses (Brühlmann et al., 2020). As shown in Table 1, the majority of the MFC studies published since 2015 (22 studies in total) have used student or online samples for the purpose of test development or empirical research. This observation highlights a potential concern regarding careless responses and the psychometric properties of the data collected.

In addition, individual differences across samples may also influence susceptibility to careless responses (Ward & Meade, 2023). For example, research suggests that individuals with lower levels of Conscientiousness, Agreeableness, Extraversion, and Emotional stability may be more likely to provide careless responses (Bowling et al., 2016). Furthermore, given that MFC measures are recognized to be more cognitively demanding than Likert-type measures (Sass et al., 2020), it is reasonable to expect that individuals with lower cognitive ability may exhibit careless responses more frequently. This is primarily because they may encounter greater challenges when engaging in the comparative judgment process compared to individuals with higher cognitive ability (e.g., Brown & Maydeu-Olivares, 2011; Lee et al., 2021; Vasilopoulos et al., 2006).

¹ Ward & Meade (2023) discussed a “survey design” as a potential source of careless responses. This study applies it to a “test design” at a test level.

Table 1 Summary Table of Recent MFC Literature

References	Construct	Format	Model	Sample	Sample context
Anguiano et al. (2015)	Emotional Intelligence	Triplet	TIRT	Online sample (n = 486)	Research Context
Lee et al. (2018)	Big Five Personality	Triplet	TIRT	Student sample (n = 417)	Research Context
Guenole et al. (2018)	Maladaptive Personality	Triplet	TIRT	Online sample (n = 420)	Research Context
Kreitchmann et al. (2019)	Big Five Personality	Pair	MUPP-2PL	Student sample (n = 609)	Research Context
Walton et al. (2020)	Big Five and HEXACO	Triplet	TIRT	Student sample (n = 5,268)	Research Context
Wetzel and Frick (2020)	Big Five Personality	Triplet	TIRT	Online sample (n = 1,286)	Research Context
Zhang et al. (2020)	Occupational Personality	Pair	MUPP	Online sample (n = 580; 806)	Research Context
Dalal et al. (2021)	15 Personality Traits	Pair	MUPP	Online sample (n = 943; 750)	Research Context
Huber et al. (2021)	Big Five Personality	Pair	MUPP	Online sample (n = 652)	Research Context
Lee et al. (2021)	Big Five Personality	Triplet	TIRT	Student sample (n = 460)	Research Context
Ng et al. (2021)	Character	Triplet	TIRT	Online sample (n = 798)	Research Context
Martínez et al. (2021)	Big Five Personality	Triplet	Quasi-Ipsative	Student sample (n = 1,141)	Research Context
Frick et al. (2021)	Big Five Personality	Triplet	TIRT	Online sample (n = 1000)	Research Context
Joo et al., (2023)	Big Five Personality	Pair	ZG-MUPP	Online sample (n = 876)	Research Context
Pavlov et al. (2021)	Big Five Personality	Pair	Not provided	Student sample (n = 516)	Research Context
Frick (2022)	Big Five Personality	Triplet	Faking Mixture	Online & Lab (n = 1,244)	Research Context
Lin (2022)	Workplace Competencies	Triplet	TIRT model	Online sample (n = 288; 349; 432)	Research Context
Martínez et al. (2022)	Big Five Personality	Triplet	Quasi-Ipsative	Student sample (n = 246)	Research Context
Lee (2023)	Big Five Personality	Triplet	TIRT model	Incumbent sample (n = 396)	Research Context
Lin et al. (2023)	HEXACO Personality	Pair	TIRT model	Online sample (n = 1,440)	Research Context
Rasheed and Robie (2023)	Vocational Interest	Triplet	Quasi-Ipsative	Student sample (n = 352 & 426)	Research Context
Borman et al. (2023)	Big Five Personality	Triplet	Not provided	Online sample (n = 1,565)	Research Context

MUPP-2PL model (Morillo et al., 2016); ZG-MUPP model (Joo et al., 2023)

Characteristics of the Context

The situational contexts in which respondents take a test or survey can be another potential source of careless responses (Meade & Craig, 2012; Ward & Meade, 2023). In high-stakes situations, careless responses are less likely to occur because test-takers are motivated to achieve high scores and thus they tend to pay more attention to the items presented. However, as shown in Table 1, the majority of research on MFC measures has been conducted in a research context rather than a real personnel selection context. In these research-focused situations, respondents may not have the same level of motivation, which consequently increases the likelihood of careless responses.

Moreover, the development of MFC measures is commonly undertaken by researchers in a research context. In the process of developing MFC tests, a critical step is to match statements within an item block based on their social desirability to minimize the risk of faking. However, if these statements within a block are closely matched in terms of social desirability, participants who lack motivation in research contexts may not exert their best efforts to rank or select the statements. Instead, they may respond arbitrarily because all options within a block seem equally positive or negative. Fuechtenhans and Brown (2022) have

acknowledged this concern, suggesting that “in the case of closely matched blocks, given a certain degree of motivation, individuals could either *select ranks carelessly* ...” (p.4).

In summary, careless responses can be caused by a variety of factors, resulting in MFC responses that do not accurately reflect the true latent traits of respondents. This, in turn, can lead to inaccuracies and biases in studies using MFC measures. To ensure the quality of research findings and organizational decision-making, researchers must use the highest quality data in test development and empirical research. In this study, we propose a method for detecting careless respondents using the I_z person-fit statistic in conjunction with the Thurstonian item response theory (TIRT; Brown & Maydeu-Olivares, 2011), which can serve as a valuable measurement tool for researchers and practitioners seeking to effectively address the issue of careless response.

Application of the I_z Person-Fit Statistic to the TIRT Model

To date, multiple IRT models have been proposed for estimating MFC data. Our research focuses on the TIRT model for two reasons. First, as shown in Table 1, the TIRT model

has been the most widely used in applied research settings. Second, researchers and practitioners can easily estimate MFC data using the Mplus program (Muthén and Muthén, 2022) or the R package (*thurstonianIRT*, Bürkner, 2018). Hence, our method can be readily accessible to researchers and practitioners who use the TIRT model to analyze MFC data.

Thurstonian IRT Model

Brown and Maydeu-Olivares (2011) proposed the TIRT model, which can be applied to different types of forced-choice formats: “more like me” associated with pairwise preference items, “most like me”, “least like me”, and “rank-order” judgments associated with three or more statements within a block.

The TIRT model is based on Thurstone’s (1927) law of comparative judgment. When comparing two statements (j and k), statement j is considered preferable to statement k if the latent utility associated with statement j (t_j) is greater than the latent utility of a statement k (t_k). This is expressed as follows:

$$y_{jk}^* = 0 \text{ if } y_{jk}^* t_j - t_k < 0, \text{ and } y_{jk}^* = 1 \text{ if } y_{jk}^* = t_j - t_k \geq 0 \quad (1)$$

where y_{jk} represents the binary response outcomes (i.e., selecting statement j over statement k) and y_{jk}^* denotes the difference in psychological values of the two statements j and k .

In the TIRT model, the latent utilities are linked to psychological attributes through the standard factor analysis model. Thus, t_j and t_k can be represented as $t_j = \mu_j + \lambda_j \eta_a + \varepsilon_j$ and $t_k = \mu_k + \lambda_k \eta_b + \varepsilon_k$. The Eq. (1) above can be extended as follows:

$$y_{jk}^* = t_j - t_k = (\mu_j - \mu_k) + (\lambda_j \eta_a - \lambda_k \eta_b) + (\varepsilon_j - \varepsilon_k) \quad (2)$$

Here, η_a and η_b represent the measured attributes a and b (i.e., factor scores), λ_j and λ_k indicates the factor loadings associated with statements j and k , and ε_j and ε_k are measurement errors. The model simplifies the mean difference between the two statements involved in a paired comparison, $(\mu_j - \mu_k)$, as a single threshold parameter (γ). The mean difference is redefined as $(\mu_j - \mu_k) = -\gamma$. As a result, each binary outcome of paired comparison is modeled by the Eqs. (3) and (4).

$$y_{jk} = 0 (k \text{ is preferred to } j) \text{ if } y_{jk}^* = -\gamma + (\lambda_j \eta_a - \lambda_k \eta_b) + (\varepsilon_j - \varepsilon_k) < 0, \text{ and} \quad (3)$$

$$y_{jk} = 1 (j \text{ is preferred to } k) \text{ if } y_{jk}^* = -\gamma + (\lambda_j \eta_a - \lambda_k \eta_b) + (\varepsilon_j - \varepsilon_k) \geq 0, \text{ and} \quad (4)$$

Because the pairwise comparison outcome is binary and the attributes are continuous, applying Thurstone’s model to the MFC measure results in an IRT model.

To describe how MFC rank responses are analyzed by the TIRT model, we revisit the triplet example introduced earlier.

Rank the three statements from “most like me (1)” to “least like me (3)”

- A. *I do things according to a plan.* 1
- B. *I make friends easily.* 2
- C. *I enjoy hearing new ideas.* 3

In this example, the ranked responses are coded as [1, 2, 3]. For the TIRT model, the ranked responses must be transformed into sets of binary responses. In the given example, the binary scoring process involves comparing different statements within an item block. When comparing statement A to statement B, a binary score of 1 is assigned because statement A is preferred over statement B. Conversely, if statement B is preferred, the binary score

would be 0. Next, a comparison is made between statement A and statement C. Since statement A is still preferred over statement C (i.e., is still ranked higher), the binary outcome of the comparison is 1. Finally, a comparison between statement B and statement C results in a binary score of 1 because statement B is preferred over statement C. As a result, the ranking response [1, 2, 3] is transformed into three sets of paired binary outcomes: [1], [1], [1].

The transformed binary outcomes (e.g., [1], [1], [1]) are then entered into the two-dimensional standard normal ogive IRT model. The item response function (IRF) of the binary outcome variable is expressed as follows:

$$P(y_{jk} = 1 | \eta_a, \eta_b) = \Phi \left(\frac{-\gamma + \lambda_j \eta_a - \lambda_k \eta_b}{\sqrt{\psi_j^2 + \psi_k^2}} \right) \quad (5)$$

where $\Phi(x)$ denotes the cumulative standard normal probability function evaluated at x . ψ_j^2 and ψ_k^2 denote residual variance for statements j and k . Equation (5) represents the conditional probability of choosing statement j over

statement k between two statements, j and k , measuring dimensions a and b (Brown, 2015).

Thus, the TIRT model consists of five parameters associated with each binary outcome of the paired comparison: two loadings (λ_j, λ_k) representing item discrimination, one threshold (γ) representing the mean difference between the two statements being compared, and two residual variances (ψ_j^2, ψ_k^2). It is important to note that the TIRT model applies equality constraints to factor loadings that correspond to the same statement during the estimation process. For instance, when comparing statement A to both statement B and statement C within a block (i.e., statement A vs. statement B, statement A vs. statement C), the factor loadings associated with statement A are constrained to be equal in both sets of comparisons. This constraint ensures that the measurement of the ranking responses remains consistent across different pairwise comparisons involving the same statement.

I_z Person-Fit

Person-fit is a model-based method used to detect misfitting respondents whose responses do not fit the IRT model (Meijer & Sijtsma, 2001; Schmitt et al., 1999). Overall, the efficacy of person-fit statistics in identifying careless respondents has been well-established through both simulation and empirical studies (Beck et al., 2019; Meijer et al., 2016; Niessen et al., 2016; Patton et al., 2019; Schneider et al., 2018). For example, Beck et al. (2019) conducted a simulation study to evaluate the effectiveness of various person-fit statistics, including U3 (van der Flier, 1982), HT, and I_z (Drasgow et al., 1985). Their results showed that the I_z person-fit statistic showed the most significant improvement in model fit after removing identified careless respondents in the simulated context. Similarly, Patton et al. (2019) conducted a simulation study that focused on the performance of the I_z person-fit statistic on model parameters. Their results showed that item parameter estimation and standard errors were significantly improved after removing careless respondents identified by the I_z person-fit statistic. Furthermore, Niessen et al. (2016) applied person-fit statistics, including the Guttman error (Meijer et al., 1994) and the I_z person-fit statistic, to data collected through a web-based survey. They concluded that “person-fit statistics are a good alternative as compared to other methods” (p.1). More recently, Schneider et al. (2018) evaluated the efficacy of person-fit statistics, including Guttman errors, normed Guttman statistics, and generalized U3, for internet-based self-report assessments. Their conclusion was that “person-fit statistics significantly and moderately distinguished between non-careless and careless responders” (p. 1077). These studies provide compelling evidence that person-fit statistics serve as effective

psychometric tools for identifying careless respondents in survey data.

Among a variety of person-fit statistics (see Karabatsos, 2003), our research focuses on the I_z person-fit statistic (Drasgow et al., 1985) which is one of the most widely used model-based person-fit statistics. The I_z person-fit statistic has been proven to be effective in detecting a variety of aberrant responses on cognitive and non-cognitive measures (e.g., Conijn et al., 2014; Drasgow et al., 1987; Ferrando, 2004; Ferrando & Chico, 2001; Li & Olejnik, 1997; Meijer & Tendeiro, 2012; Nering, 1997; Nering & Meijer, 1998; Reise & Flannery, 1996; Schmitt et al., 1999; Tendeiro et al., 2016; Zickar and Drasgow, 1996). The I_z person-fit statistic can be applied to the TIRT model as follows: suppose n is the total number of MFC binary outcomes for the TIRT model. The log-likelihood l_0 of an individual's response pattern is given by

$$l_0 = \sum_{i=1}^n y_i \log[P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b)] + (1 - y_i) \log[1 - P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b)] \quad (6)$$

The estimated latent traits ($\hat{\eta}_a, \hat{\eta}_b$) and estimated statement parameters of the TIRT model are used to compute the log-likelihood in the Eq. (6). The expectation of the log-likelihood $E(l_0)$ is shown as follows:

$$E(l_0) = \sum_{i=1}^n P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b) \log[P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b)] + [1 - P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b)] \log[1 - P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b)] \quad (7)$$

The approximate variance is

$$Var(l_0) = \sum_{i=1}^n P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b) [1 - P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b)] \left\{ \log \frac{P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b)}{1 - P(y_i = 1|\hat{\eta}_a, \hat{\eta}_b)} \right\}^2 \quad (8)$$

Finally, the I_z person-fit statistic of the TIRT model is represented by

$$I_z = \frac{l_0 - E(l_0)}{\sqrt{Var(l_0)}} \quad (9)$$

To assess person misfit, I_z values obtained from Eq. (9) are compared to a critical value obtained from a standard normal distribution or an empirical distribution. If the observed I_z value is less than the critical value, the response pattern of an examinee is identified as a *person misfit*.

I_z values are often assumed to be standardized and normally distributed, but research has found that they do not exactly follow a standardized normal distribution (Drasgow et al., 1985; Molenaar & Hoijtink, 1990). The distribution of I_z can deviate from normality when estimated latent trait parameters are used to calculate I_z person-fit statistics (Beck et al., 2019). Thus, the theoretical critical value of I_z , such as -1.64 (corresponding to the 5% quantile of the normal

distribution), can distort Type I error and power rates (Lee et al., 2014; Molenaar & Hoijtink, 1990; Nering, 1997; Snijders, 2001). To address this issue, researchers have proposed using an empirical critical value instead of a theoretical critical value in various contexts (Armstrong et al., 2007; Cui & Li, 2015; Lee et al., 2014; Nering, 1997). A two-step approach is employed to determine the empirical critical value. First, usual (e.g., normal) response patterns are simulated to represent the expected responses under honest test conditions, aligning with the actual scale characteristics. Then, the I_z values are calculated for the generated samples that exhibit these normal response patterns. This results in an empirical distribution of I_z values. The value corresponding to the 5% quantile of the distribution is identified as the empirical critical value. Finally, this empirical critical value is compared to the observed I_z values obtained from the real data.

The Present Study

In this article, we conducted two studies. While prior research extensively documented the prevalence of careless responses on Likert-type measures, there is a notable scarcity of empirical evidence regarding their occurrence on MFC measures. Therefore, as an initial research endeavor, we sought to investigate whether careless responses are as common on MFC measures as they are on Likert-type scales. To address this, we formulated the following research question based on a within-subject design:

Research Question 1 (RQ1) Are careless responses as common on MFC measures as they are on Likert-type scales?

In Study 2, we performed Monte Carlo simulation studies to evaluate the effectiveness of our proposed method. Simulation studies allow researchers to investigate the extent to which various conditions affect statistical results and shed light on various issues that cannot be fully observed in real-world data (e.g., Leite et al., 2022). For this reason, simulation studies have been often used to investigate the effects of careless responses (e.g., Beck et al., 2019; DeSimone et al., 2018; Meade & Craig, 2012; Niessen et al., 2016; Patton et al., 2019; Schroeders et al., 2022). We conducted simulation studies under six different conditions. We also examined whether removing identified careless cases (i.e., person-misfit) improves reliability and criterion-related validity.

Since our research proposes a new methodological approach for identifying the careless respondents in MFC data, it is important to demonstrate that our method produces low Type I error (i.e., false-positive rate) and sufficient power (i.e., true-positive rate) results. Type I error rate refers to the probability of misidentifying respondents as careless respondents when they are not in fact careless. On the other

hand, the power rate refers to the probability of correctly identifying truly careless respondents. In the research context of detecting careless respondents, maintaining a low Type I error is particularly important. Regardless of how high the power value may be, if the Type I error associated with misclassification is not adequately controlled, the effectiveness of the proposed methodology will be substantially limited. Therefore, it is crucial to show that Type I errors are well-controlled to ensure that individuals are not misclassified as careless respondents. For this reason, we first investigate the Type I error of our proposed method and then examine its power. By conducting a systematic investigation of both Type I error and power, our research provides a thorough evaluation of the effectiveness of our proposed method in simulation.

Research Questions 2 (RQ2) How well does the proposed I_z person-fit method for the TIRT model control Type I error?

Research Questions 3 (RQ3) How accurately does the proposed I_z person-fit method detect careless respondents to be careless respondents?

After assessing the efficacy of person-fit statistics, it is important to examine whether the person-fit statistics can truly enhance the psychometric properties of response data (e.g., Beck et al., 2019; Patton et al., 2019). To accomplish this goal, we formulated two research questions to examine potential improvements in reliability and criterion-related validity resulting from the removal of identified careless respondents. By examining the improvements in reliability and criterion-related validity, we aim to provide evidence of the practical utility of our proposed method.

Research Question 4 (RQ4) How much does the reliability improve after removing the identified person-misfit cases?

Research Question 5 (RQ5) How much does the criterion-related validity improve after removing the identified person-misfit cases?

Study 1

Sample and Measures

We recruited 306 full-time undergraduate students from a large university in the US through the university's Psychology Research Participation System. Participants were asked to complete two sets of personality measures through an online survey. To do this, we used the 20-triplet maladaptive personality MFC measure developed by Guenole et al. (2018). This measure consists of 20-triplet measures for six psychological constructs (i.e., negative affect, detachment,

antagonism, disinhibition, compulsivity, and psychoticism) according to the DSM-5 maladaptive model. The statements were matched on a similar level of social desirability and each trait was measured by 12 statements. An example MFC item is as follows:

Rank the three statements in each group from “most like me (1)” to “least like me (3)”

- (A) *Insist on everything being perfect.*
- (B) *Get into trouble.*
- (C) *Feel miserable.*

Each statement was initially validated by Guenole et al. (2018) first, and then the authors created MFC items by composing the statements. Thus, the same statements from the maladaptive personality MFC measures were also used for the Likert-type measures, for which participants responded on a five-point agreement scale ranging from “Strongly disagree” (1) to “Strongly agree” (5).

We employed a within-subject design in which participants completed the first set of personality measures at Time 1 and then to the second set of personality measures at Time 2 (i.e., two weeks later). Participants were randomly assigned into two groups. Half of the participants completed the MFC measure of maladaptive personality first, while the other half completed the Likert-type measure first to minimize order effects. After completing each personality measure, participants responded to a self-reported single-item indicator. Self-reported single-item indicators have been successfully used in previous studies to evaluate careless responding behaviors (Meade & Craig, 2012; Ward & Meade, 2018). For example, Meade and Craig (2012) included “a series of single-item measures at the end of the survey intended to allow respondents to indicate how much effort and attention they devoted to the study (p. 5)”. In our study, we adapted the example of Meade and Craig’s (2012) as follows:

In this study, we aim to examine the impact of careless responses on the quality of data. We need your honest opinion. Your answer would not affect your participation credit. That is, you will receive credit for this study no matter what. In your honest opinion, how often did you carelessly respond to questions in the personality test?

Participants responded to these items on a five-point frequency scale: 1 = *never*, 2 = *rarely*, 3 = *sometimes*, 4 = *often*, 5 = *always*.

We also included questions about perceived cognitive load and test fatigue to evaluate the validity of the MFC data we collected. Although these measures are not directly related to RQ1, they can help us evaluate whether test-takers’ reactions to our MFC data are consistent with previous

research findings that MFC measures are perceived to be more cognitively demanding and fatiguing (e.g., Bowen et al., 2002; Dalal et al., 2021; Sass et al., 2020). This can serve as a validity check of the MFC data we collected. We used six items to measure perceived cognitive load (Hart & Staveland, 1988), with an example item, “I spent a large amount of mental effort completing this personality test.” We measured test fatigue using a three-item measure (Speer et al., 2016) with an example item, “This test wore me out.” The internal consistency indices of the cognitive load and test fatigue measures were 0.78 and 0.79, respectively. Both measures used five-point Likert scales (e.g., strongly disagree = 1 to strongly agree = 5).

Students were asked to register for Time 1 and Time 2 sessions separately, but 51 students did not register for the second session. Therefore, 255 students remained in our final sample. The final sample was primarily female (67%). The mean age of the participants was 20.91 years with a standard deviation of 4.13 years. Regarding racial identity, 42% of participants identified as Caucasian, 21% as Asian, 15% as Hispanic, and 14% as Black. The remaining 8% identified with some other racial identity.

Analysis

Paired samples t-tests were conducted, and Cohen’s *d*-effect size was calculated. The criteria for Cohen’s effect size (Cohen, 1998) were as follows: 0.2 (small effect), 0.5 (moderate effect), and 0.8 (large effect).

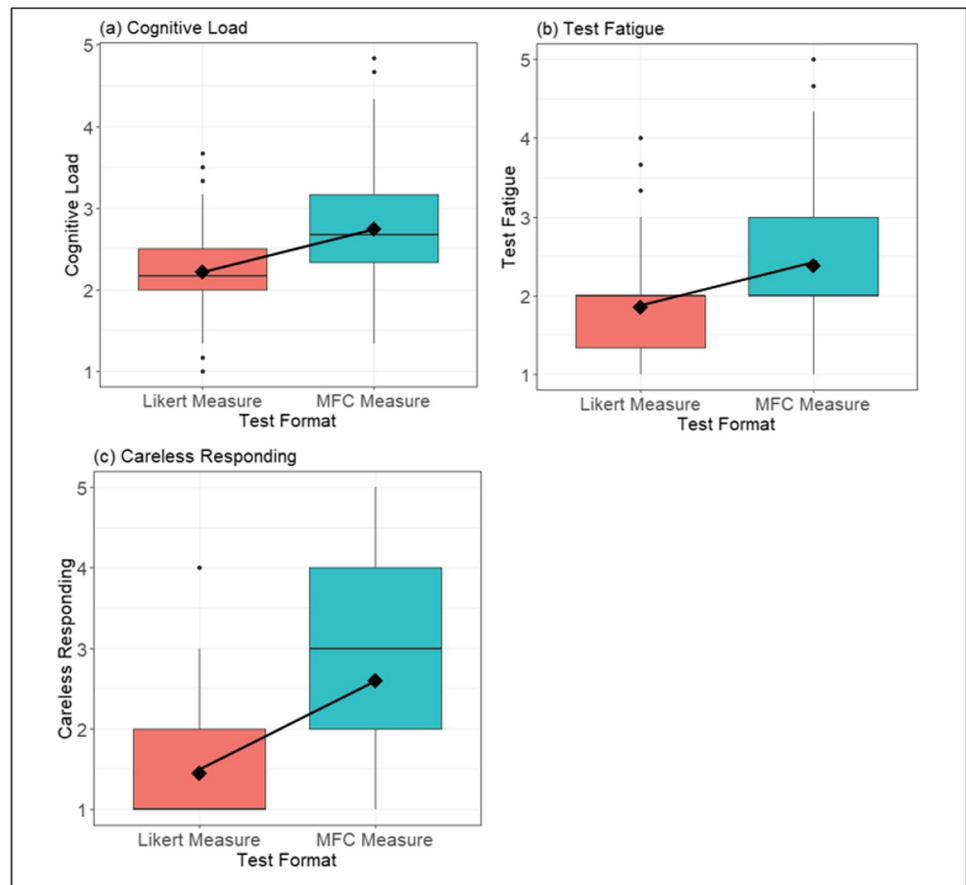
Results

RQ 1: Are careless responses as common on MFC measures as they are on Likert-type scales?

To ensure the validity of our response data collected through the MFC measure, we first evaluated participants’ perceptions of cognitive load and test fatigue. Results indicated that participants experienced higher levels of cognitive load and test fatigue when responding to the MFC measure of maladaptive personality than to the corresponding Likert-type measure. The effect sizes (*d*) were moderate, with values of 0.67 and 0.54, respectively: $t(509) = 70.47$, $p < 0.001$, $M_{\text{MFC}} = 2.74$, $SD_{\text{MFC}} = 0.68$, $M_{\text{Likert}} = 2.22$, $SD_{\text{Likert}} = 0.48$ for cognitive load and $t(509) = 44.71$, $p < 0.001$, $M_{\text{MFC}} = 2.38$, $SD_{\text{MFC}} = 0.93$, $M_{\text{Likert}} = 1.86$, $SD_{\text{Likert}} = 0.61$ for test fatigue. These results were generally consistent with previous findings that suggest that MFC measures are perceived as more cognitively demanding and fatiguing than Likert-type measures (e.g., Bowen et al., 2002; Dalal et al., 2021; Sass et al., 2020).

RQ1 examines whether careless responses are as common on the MFC measure as they are on the Likert-type

Fig. 1 Boxplot of Test-Takers' Reaction Outcomes across Test Formats. Note. Diamond in the box indicates "mean" for each measure. Averaged scores were used for cognitive load and test fatigue



scale. Our results indicate that respondents are at least as likely or more likely to respond carelessly on the MFC measure than on the Likert-type scale: $t(509) = 88.04$, $p < 0.001$, $M_{\text{MFC}} = 2.60$, $SD_{\text{MFC}} = 1.18$, $M_{\text{Likert}} = 1.45$, $SD_{\text{Likert}} = 0.67$. Figure 1 provides a visual representation of the overall pattern concerning cognitive load, test fatigue, and careless responding.

Study 2

Simulation Design

To explore the performance and effectiveness of the I_z statistic for the TIRT model, we conducted the Monte Carlo simulation study under various conditions. The number of latent trait dimensions was set to five, corresponding to the Big Five personality, and the sample size was fixed at 1,000 following previous simulation studies (e.g., Brown & Maydeu-Olivares, 2011; Joo et al., 2020; Lee et al., 2019a, 2019b). A total of 60 statements were used to create 30-pair and 20-triplet measures using a balanced design of 12 statements per dimension. Six simulation conditions were considered as follows:

- (1) **Types of MFC data:** As shown in Table 1, most MFC research has used triplet format, while pair format has also been used in practice (e.g., Joo et al., 2023; Kreitchmann et al., 2019; Stark et al., 2005). Therefore, both *pair* and *triplet* MFC data were considered for the simulation conditions based on their use in research and applied settings.
- (2) **Proportion (%) of respondents answered carelessly:** Three different levels of prevalence of careless respondents were considered based on previous empirical studies (e.g., Curran, 2016; Goldammer et al., 2020; Johnson, 2005; Meade & Craig, 2012; Oppenheimer et al., 2009): (i) 10%, (ii) 20%, (iii) 50%.
- (3) **Proportion (%) of items answered carelessly:** Respondents may produce different numbers of careless responses. Previous simulation studies have investigated the performance of person-fit under various proportions of carelessly answered items, such as 4%, 10%, 20%, and 40% (Levine & Rubin, 1979); 15% and 30% (Drasgow et al., 1987); 20% and 30% (Nering & Meijer, 1998); 20%, 40%, and 50% (Sinharay, 2017). In this study, four conditions were considered: (i) 20%, (ii) 30%, (iii) 50%, and (iii) 100%. The 100% condition represents situations in which respondents are completely inattentive and unmotivated. It may be unusual

to observe such a high proportion of careless responses in the real world. Nevertheless, by including this condition in the simulation study, we aim to investigate the effectiveness of our method under exceptionally challenging circumstances and to evaluate the upper limits of the careless response scenario.

- (4) **Types of item parameters (a.k.a. statement parameters):** In a real-world situation, the presence of careless responses may negatively influence the estimation of TIRT item parameters, which may, in turn, affect the performance of the I_z person-fit statistic (e.g., Sinharay, 2017). To investigate how item parameters affect the performance of the I_z person-fit statistic on MFC data, two types of item parameters were used to compute the I_z statistic: (i) *estimated* and (ii) *true* TIRT item parameters. Estimated item parameters were obtained from the response data, including careless responses. True item parameters were predetermined from the simulation conditions without estimating response data. In operational settings, the true IRT item parameters can be considered as item parameters from pre-tests or previous administrations. In research settings, they can be considered as item parameters from published research.
- (5) **I_z critical values:** Both (i) *empirical* and (ii) *theoretical* critical values were used to identify the careless respondents. To obtain the empirical critical value, "normal" data was first generated based on true item parameters with 1000 samples (a.k.a. simulees) from the multivariate normal distribution. Item parameters and latent trait scores of the TIRT model were then estimated, and the I_z values were calculated for each simulee using the estimated parameters from the TIRT, rather than the population true parameters. The resulting I_z values were sorted in ascending order, and a lower one-tailed value corresponding to the 5th percentile of the I_z distribution was identified and used as the empirical I_z critical value. We note that the empirical critical values were determined separately for pair and triplet MFC measures. A value of -1.64, which corresponds to the 5th percentile of the standard normal distribution, was used for the theoretical critical value.
- (6) **Latent trait correlations:** We also evaluated how latent trait correlations affect the performance of the I_z person-fit statistic on MFC data. Based on previous simulation research, two different theta correlations were considered: (i) 0 and (ii) 0.5 (e.g., Brown & Maydeu-Olivares, 2011; Joo et al., 2018, 2020; Lee et al., 2019a, 2019b).
- (1) Five latent trait scores were sampled from a multivariate normal distribution, $MVN(0, \Sigma)$, with the covariances among the dimensions set to 0 and 0.5 depending on conditions. Sixty measurement errors (ψ_j^2) were also randomly drawn from a univariate standard normal distribution. The criterion variable was also simulated by correlating with five latent trait scores at 0.5 for evaluating the validity improvement.
- (2) True item parameters were randomly generated from uniform distributions $U(0.5, 2)$ for factor loadings and $U(-2, 2)$ for thresholds, respectively.
- (3) "Normal" binary outcomes for MFC pair and triplet measures were generated based on the TIRT model using the true latent trait scores and item parameters.
- (4) "Careless" responses were separately generated as follows. For the pair format, binary outcomes were randomly selected from 0 or 1. For the triplet format, binary outcomes were randomly selected from six possible responses ([1,1,1], [1,1,0], [0,1,1], [1,0,0], [0,0,0], [0,0,1]). The normal responses were then replaced with careless responses depending on the simulation conditions.

A total of 192 experimental conditions were created ($2 \times 3 \times 4 \times 2 \times 2 \times 2$) and 100 replications were conducted for each condition. The data generation, TIRT model estimation, and I_z person-fit method were simultaneously conducted using Mplus 8.0 and the R program.

Performance Evaluation

To identify careless respondents, the observed I_z values were calculated using estimated and true item parameters in combination with estimated latent trait scores. Simulees were classified as "misfit" when the observed I_z values were less than the empirical or theoretical critical I_z values. Type I error and power rates were computed across replications for each condition. Type I error rates were calculated as the proportions of normal respondents incorrectly identified as misfit across replications. Similarly, power rates were calculated as the proportions of careless respondents are correctly identified as misfit across replications. The correlations between the criterion variable and the five latent trait scores were examined with and without excluding classified person-misfit cases. The validity improvement was computed by $(\text{validity with misfit removal} \div \text{validity without misfit removal} - 1) \times 100$. The results were then arithmetically averaged across all dimensions. To evaluate the reliability improvement, we calculated the empirical reliability: $r_{(\hat{\theta}, \theta)}^2$.

Pearson correlations were calculated between the generating θ parameters and the estimated scores, namely the $\hat{\theta}$

Generating Careless Response Data

Careless responses and criterion variables were generated as following steps:

Table 2 Type I Error Rates for the l_z Person-fit Statistic on Pair and Triplet MFC Measures

Cor	% Careless respondents	% Careless Items	Pair MFC format				Triplet MFC format			
			Estimated item parameters		True item parameters		Estimated item parameters		True item parameters	
			Empirical	Theoretical	Empirical	Theoretical	Empirical	Theoretical	Empirical	Theoretical
0	10	20	.04	.00	.05	.01	.03	.00	.05	.00
		30	.03	.00	.05	.01	.03	.01	.05	.00
		50	.03	.00	.05	.01	.02	.00	.05	.00
		100	.02	.00	.05	.01	.01	.00	.05	.00
	20	20	.04	.00	.05	.01	.03	.00	.05	.00
		30	.03	.00	.05	.01	.02	.01	.05	.00
		50	.02	.00	.05	.01	.01	.00	.05	.00
		100	.01	.00	.05	.01	.00	.00	.05	.00
	50	20	.03	.00	.05	.01	.03	.00	.05	.00
		30	.02	.00	.05	.00	.03	.00	.05	.00
		50	.01	.00	.05	.01	.01	.00	.05	.00
		100	.00	.00	.05	.01	.00	.00	.05	.00
0.5	10	20	.05	.00	.05	.01	.04	.00	.05	.00
		30	.04	.00	.05	.01	.03	.00	.05	.00
		50	.04	.00	.05	.01	.02	.00	.05	.00
		100	.03	.00	.05	.01	.01	.00	.05	.00
	20	20	.05	.00	.05	.01	.03	.00	.05	.00
		30	.04	.00	.05	.00	.02	.00	.05	.00
		50	.03	.00	.05	.01	.01	.00	.05	.00
		100	.02	.00	.05	.01	.00	.00	.05	.00
	50	20	.04	.00	.05	.01	.03	.00	.05	.00
		30	.04	.00	.05	.01	.02	.00	.05	.00
		50	.01	.00	.05	.01	.01	.00	.05	.00
		100	.00	.00	.05	.01	.00	.00	.05	.00

parameters obtained by the TIRT method. The correlations were then squared and averaged across replications. We note that this reliability ($r^2_{(\hat{\theta}, \theta)}$) has been often reported in previous MFC studies (Brown & Maydeu-Olivares, 2011; Burkner, 2018; Lee et al., 2022; Lin, 2022). The empirical reliabilities were calculated with or without the elimination of identified person-misfit cases. We took an average of the reliabilities across five dimensions for the summary. The reliability improvement was also computed by (reliability with misfit removal $\div$ reliability without misfit removal $- 1$) $\times 100$.

Results

RQ2: How well does the proposed l_z person-fit method for the TIRT model control Type I error?

Table 2 shows the Type I error rates of the l_z person-fit statistic for pair and triplet MFC measures under various

conditions. Type I error rates were very well-controlled regardless of response type, test format, theta correlations, percentages of careless items and respondents, type of item parameters, and type of critical values. Specifically, most Type I error rates were similar to or less than the nominal alpha level of 0.05, indicating that the proposed l_z person-fit statistic of the TIRT model can effectively control misclassification and provide valid results.

RQ3: How accurately does the proposed l_z person-fit method identify careless respondents to be careless respondents?

Table 3 shows the power rates of the l_z person-fit statistic for pair and triplet MFC measures under various conditions. First, as the proportion of careless items increases, so do the power rates. For example, in the condition of $r=0$, 50% careless respondents, true item parameters, empirical l_z critical value, and triplet format, the power rates increased from

Table 3 Power Rates for the I_z Person-fit Statistic on Pair and Triplet MFC Measures

Cor	% Careless respondents	% Careless items	Pair MFC format						Triplet MFC format	
			Estimated item parameters		True item parameters		Estimated item parameters		True item parameters	
			Empirical	Theoretical	Empirical	Theoretical	Empirical	Theoretical	Empirical	Theoretical
0	10	20	.15	.00	.14	.05	.19	.01	.21	.01
		30	.28	.01	.33	.15	.38	.05	.55	.13
		50	.41	.02	.52	.24	.58	.09	.77	.25
		100	.60	.07	.71	.43	.92	.50	.97	.71
	20	20	.11	.00	.19	.05	.10	.00	.22	.01
		30	.13	.01	.35	.13	.29	.04	.60	.18
		50	.31	.01	.51	.24	.42	.04	.77	.24
		100	.48	.03	.70	.41	.83	.31	.97	.71
	50	20	.06	.00	.20	.05	.06	.00	.21	.01
		30	.09	.00	.41	.12	.08	.01	.48	.13
		50	.11	.00	.51	.23	.10	.00	.77	.24
		100	.16	.00	.70	.41	.35	.03	.97	.72
0.5	10	20	.18	.00	.21	.06	.17	.01	.23	.01
		30	.26	.01	.31	.14	.39	.04	.51	.14
		50	.43	.03	.52	.25	.58	.09	.77	.23
		100	.60	.08	.70	.41	.91	.50	.97	.71
	20	20	.14	.00	.20	.05	.12	.00	.24	.01
		30	.15	.01	.34	.14	.30	.05	.58	.18
		50	.33	.02	.51	.25	.43	.04	.77	.24
		100	.48	.03	.69	.43	.81	.31	.97	.70
	50	20	.08	.00	.20	.05	.06	.00	.23	.01
		30	.09	.00	.40	.13	.08	.00	.50	.14
		50	.13	.00	.51	.25	.12	.00	.77	.24
		100	.18	.00	.69	.42	.35	.04	.97	.71

0.21 to 0.48, 0.48 to 0.77, and 0.77 to 0.97 as the proportion of careless items increased from 20 to 30% to 50% to 100%. In general, the pattern was consistent across conditions. The pattern shown in Fig. 2(a) indicates that power rates improve as the percentage of careless items increases.

Second, using the empirical I_z critical value results in greater power rates than the theoretical I_z critical value. For instance, under the condition of $r=0$, 50% careless respondents, true item parameters, and triplet format, the empirical I_z critical value had a power rate of 0.77, while the theoretical I_z critical value had a power rate of 0.24. Figure 2(b) illustrates an average power level between the two critical values, demonstrating that the empirical I_z critical value yields significantly higher power rates.

Third, the power rates were greater when true item parameters were utilized as opposed to estimated item parameters. For example, under the condition of $r=0.5$, 20% careless respondents, empirical I_z critical value, and triplet format, the power rate for the true item parameters was 0.77,

whereas it was only 0.43 for the estimated item parameters. Figure 2(c) compares the average power rates produced by the two sets of item parameters, revealing that the true item parameters generated significantly higher power rates than the estimated item parameters.

Fourth, the triplet format was observed to have higher power rates than the pair format. For example, under the conditions of $r=0.5$, 50% careless respondents, empirical I_z critical value, and true item parameters, the power rate for the triplet format was 0.97, while for the pair format, it was 0.69.

ANOVA was conducted to further interpret the power results. Table 4 displays the results of an ANOVA for main effects and interactions accounting for at least 1% of the variance in power. To evaluate the effect size, omega-square (ω^2) was utilized. The ω^2 values of 0.01, 0.06, and 0.14, respectively, indicate small, medium, and large effects (Cohen, 1988). The results indicated that the percentage of careless items ($\omega^2 = 0.35$), the type of critical values ($\omega^2 = 0.25$), and the type of item parameters ($\omega^2 = 0.15$) all had

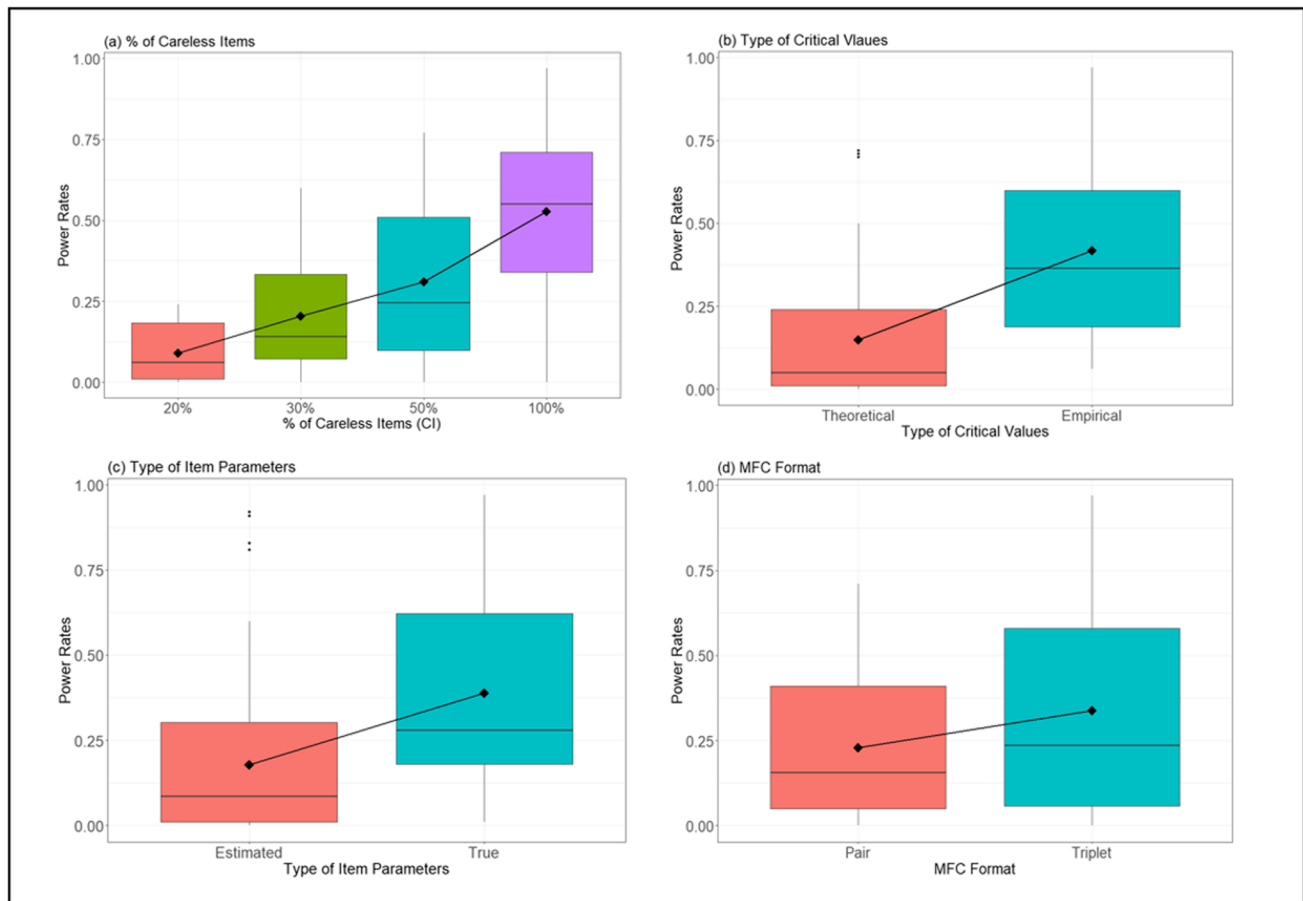


Fig. 2 Boxplot of Power Rates on % of Random Items, Critical Values, Types of Parameters, and Formats. Note. The diamond in the box indicates “mean” for each measure

Table 4 Main Effects and Interactions for Independent Variables on Power Rates

Source	df_B	F	p	ω^2
Percent of careless items (%CI): 20, 30, 50, 100%	3	419.10	<.001	0.35
Type of critical values (TC): Empirical, Theoretical	1	874.01	<.001	0.25
Type of item parameters (TI): True, Estimated	1	535.52	<.001	0.15
Type of formats (TF): Pair, Triplet	1	144.80	<.001	0.04
Percent of careless respondents (%CR): 10, 20, 50%	2	40.97	<.001	0.02
Latent correlations (LC): 0, 0.5	1	0.11	0.74	0.00
%CI*TF	2	40.39	<.001	0.04
%CI*TI	2	40.85	<.001	0.04
%CI*TC	1	11.81	<.001	0.02
%CR*TI	2	33.26	<.001	0.02

Only interaction effects that accounted for at least 1% of the variance in power are included. ω^2 = proportion of variance accounted for by the independent variables. df_B = degrees of freedom between; for all effects; degrees of freedom within = 244

large effects on power rates. The type of response formats ($\omega^2 = 0.04$) had a medium effect, while the percentage of careless respondents and latent correlations had small to negligible ($\omega^2 = 0.02$ and 0.00 , respectively). Moreover, interaction effects were small to negligible.

RQ4: How much does the reliability improve after removing the identified person-misfit cases?

We note that the RQ4 and RQ5 focused on the triplet MFC format over the pair format due to its superior performance.

Table 5 Validity and Reliability Improvement after Removing Random Respondents for Triplet MFC Measures

Theta Corr	% Careless respondents	% Careless items	Reliability improvement	Validity improvement
0	10	20	0.42%	0.76%
		30	1.22%	1.35%
		50	2.20%	2.47%
		100	6.22%	6.46%
	20	20	0.59%	1.12%
		30	2.10%	2.89%
		50	4.19%	4.66%
		100	13.52%	14.05%
	50	20	0.94%	1.31%
		30	5.86%	6.22%
		50	10.45%	10.42%
		100	50.10%	50.38%
0.5	10	20	0.49%	2.70%
		30	1.25%	3.14%
		50	2.29%	4.45%
		100	6.34%	8.38%
	20	20	0.69%	2.87%
		30	2.24%	4.98%
		50	4.55%	6.50%
		100	13.65%	15.53%
	50	20	1.00%	3.42%
		30	6.25%	7.32%
		50	10.85%	12.30%
		100	49.77%	51.91%

Improvement = (Value with misfit removal ÷ Value without misfit removal – 1) * 100

In addition, the empirical l_c critical values were used. Table 5 shows the improvements in reliability after excluding the identified person-misfit cases. Figure 3 shows a boxplot with a mean value to facilitate interpretation. The results show that as the percentage of careless respondents increased, so did the improvements in reliability. For example, under the condition of $r=0.5$ and 50% careless respondents, the reliability improved by 1.00% (from 0.85 to 0.86), 6.25% (from 0.80 to 0.85), 10.85% (from 0.78 to 0.86), and 49.77% (from 0.59 to 0.88), respectively, as the proportion of careless items increased to 20%, 30%, 50%, and 100%. In contrast, latent correlations did not affect the improvement in validity. On average, there was an approximately 9% improvement in reliability across all conditions.

RQ5: How much does the criterion-related validity improve after removing the identified person-misfit cases?

Table 5 also shows the improvements in validity after removing identified person-misfit cases and Fig. 4 illustrates a boxplot with a mean value. The results indicate that

the improvements in validity increased as the percentage of careless items increased. For instance, under the same condition above (i.e., $r=0.5$ and 50% careless respondents), the validity improved by 3.42% (from 0.42 to 0.44), 7.32% (from 0.41 to 0.44), 12.30% (from 0.39 to 0.43), and 51.91% (from 0.29 to 0.45), respectively, as the proportion of careless items increased to 20%, 30%, 50%, and 100%. Similar to the results of reliability, latent correlations did not affect the improvement in validity. On average, there was an approximately 11% improvement in validity across all conditions.

Discussion

In recent years, MFC measures have gained increasing popularity due to their ability to reduce faking responses. However, for a variety of reasons (e.g., test design, administration, sample, and context), careless responses can occur when using MFC measures, which can introduce measurement error and bias into assessments of psychological traits and affect the quality of research findings and organizational decision-making (Lee et al., 2021). To ensure the high quality of MFC data, it is critical to address the issue of careless response. Despite the significant advances in MFC research, there has been no explicit investigation of this matter. Therefore, our study fills this research gap and provides valuable insights into how to address the issue of careless responses in MFC data. We note that R syntax and source code can be found in the appendix and supplementary materials. These materials will help researchers and practitioners apply the l_c person-fit statistic to the TIRT model in their research projects.

Main Findings and Implications

Study 1 examined whether careless responses are as common on MFC measures as they are on Likert-type measures. The results revealed that respondents were at least equally or even more prone to providing careless responses on the MFC measure than on the Likert-type measure. These findings underscore the need for a statistical methodology to identify and classify careless respondents in MFC response data, similar to the methods employed for Likert-type response data. Although we used cognitive load and test fatigue only to check the validity of the MFC data in Study 1, the result still provides some valuable implications. The results showed moderate effect sizes between MFC and Likert-type measures for cognitive load and Likert-type measures ($d=0.67$ and 0.54, respectively), indicating that MFC measures are more cognitively demanding and fatiguing than Likert-type measures. This finding has implications for the MFC test design to reduce cognitive load and test fatigue and improve

Fig. 3 Boxplot of Reliability Improvements across Simulation Conditions. Note. The diamond in the box indicates “mean” for each measure

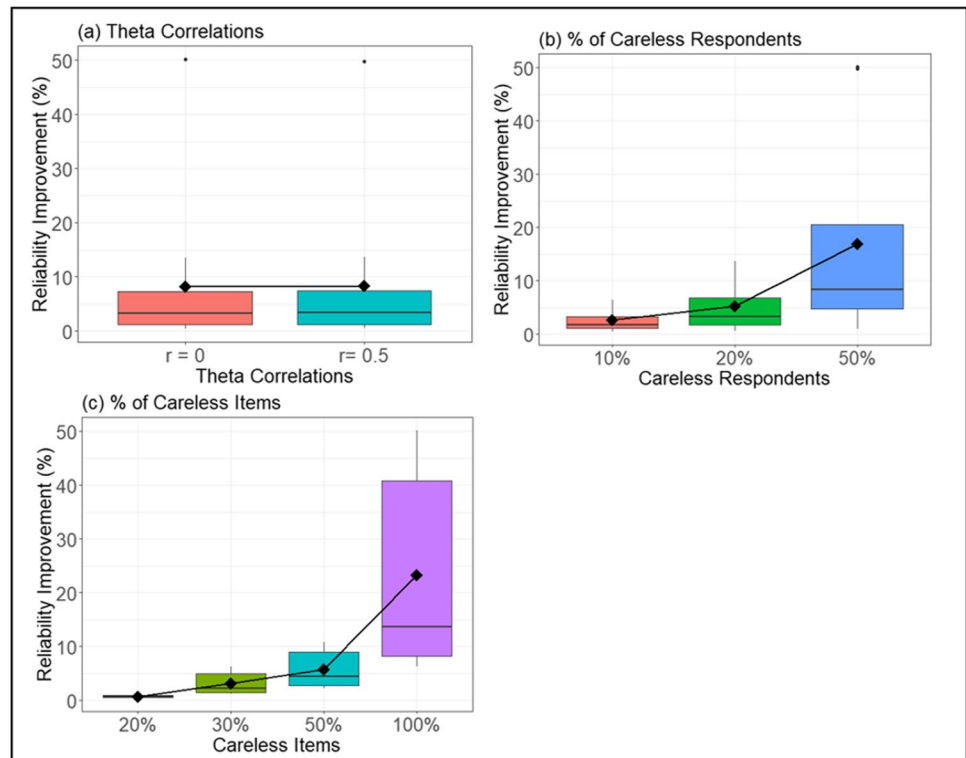
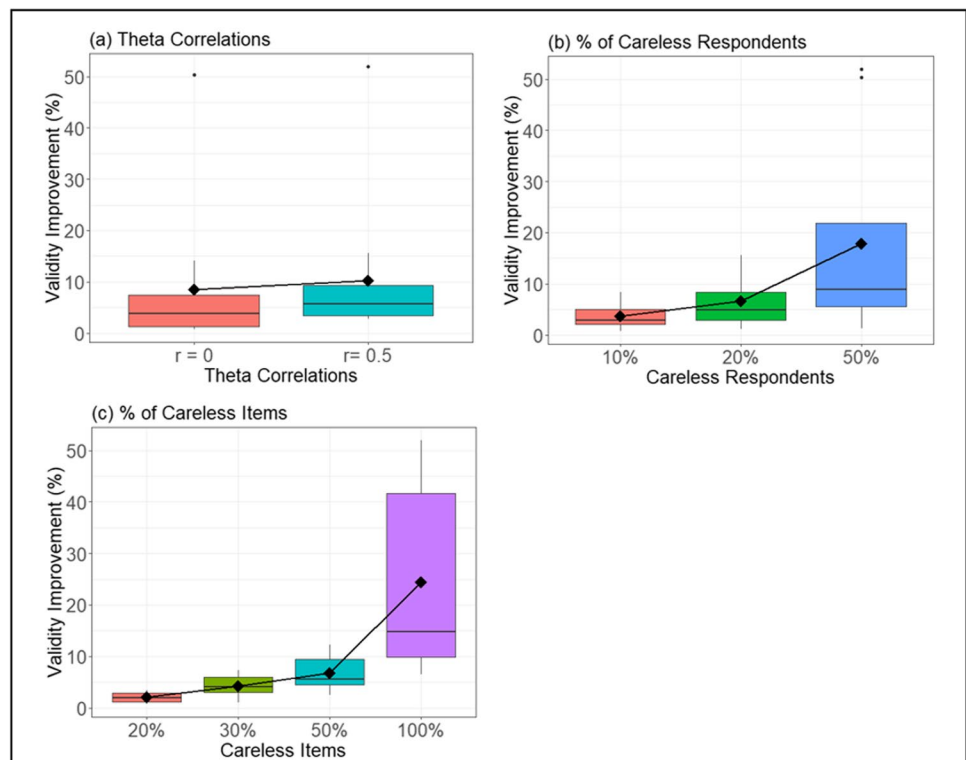


Fig. 4 Boxplot of Validity Improvements across Simulation Conditions. Note. The diamond in the box indicates “mean” for each measure



overall test-taker reactions. Lee et al. (2021) suggested that cognitive load may be higher for the “rank” format than the “pick” (i.e., asking respondents to choose only the “*most like me*” statement rather than rank them) format. Given that our MFC measure was a rank format, a simpler format may be preferable to complex formats to avoid contamination by such nuisance determinants and to reduce the adverse impact on protected groups (Lee et al., 2021). Furthermore, a graded MFC format (Dalal et al., 2021) may be an effective alternative format because it allows respondents to express the strength of their endorsement rather than simply forcing them to choose one statement over another within an item block. Dalal et al. (2021) showed that the graded pairwise preference format elicited more positive reactions (e.g., willingness to perform, appropriateness, attraction to the organization, and intentions to recommend) than the traditional pairwise preference format. These alternative approaches to the MFC format may reduce cognitive load and test fatigue, thereby improving test-takers’ reactions.

Study 2 evaluated the efficacy of our proposed method. The main findings are summarized as follows. First, our proposed method exhibits excellent control over Type I errors, a critical aspect in ensuring minimal misclassification errors. Misclassification errors can have significant consequences, including the loss of valid response data (e.g., those representing normal responses) and a reduction in the overall sample size (Arthur et al., 2021). Our proposed method can be used in research and applied settings without a significant risk of misclassification.

Second, as the proportion of careless items increased, there was a consistent increase in power rates. This finding aligns with our expectations, as an increased prevalence of disparities between observed and predicted item response probabilities facilitates the I_z person-fit statistic in effectively discerning between carefully and carelessly responded items.

Third, higher power rates were found for the triplet format than for the pair format. According to previous studies, the triplet format is more informative than the pair format, resulting in a more precise estimation of latent trait parameters (Joo et al., 2018, 2020; Lee et al., 2019a, 2019b). Our results are consistent with previous findings. That is, with the same number of statements (e.g., 60 statements), the triplet format (e.g., 20 triplets) can be more effective at detecting careless respondents than the pair format (e.g., 30 pairs). This finding would be good news for researchers and practitioners who commonly use the triplet format in their testing applications.

Fourth, higher power rates were achieved when the empirical critical value was used rather than the theoretical critical value. This result is consistent with previous research using single-statement Likert-type formats (e.g., Armstrong et al., 2007; de la Torre & Deng, 2008; Lee et al., 2014; Nering, 1997; van Krimpen-Stoop & Meijer, 2002). This

finding suggests that when detecting careless respondents in MFC data, using the empirical critical value will yield more accurate and valid results. Therefore, we recommend that researchers and practitioners use the empirical critical value when using the I_z person-fit in combination with the TIRT model.

Fifth, higher power rates were observed when the I_z person-fit statistic was calculated using true item parameters as opposed to estimated item parameters. This result is not surprising given that true parameters in simulation scenarios do not involve estimation error. However, in the real world, researchers often rely on estimating item parameters from actual response data, which may include careless responses. Consequently, estimation errors resulting from these careless responses can have a detrimental effect on the accuracy of I_z detection. Hence, it is important to use unbiased item parameters in calculating the I_z person-fit statistic. This particular concern is discussed in more detail in the *practical consideration* section below.

Sixth, our results reveal that correlations between latent traits (i.e., theta) had little effect on power rates. Thus, regardless of latent trait correlations, the I_z person-fit statistic can be used effectively to detect careless responses in MFC data.

Finally, removing careless respondents improved reliability and criterion-related validity. The greater the proportion of careless respondents and careless items, the greater the improvement in reliability and criterion-related validity. Previous research has demonstrated that removing careless respondents improves psychometric properties and reduces the risk of Type I or Type II errors in statistical inferences (Huang et al., 2015). Consistent with previous research, our empirical findings demonstrate the utility of the I_z person-fit statistic in improving the psychometric quality of MFC data.

Regarding the results, we urge special caution in interpreting the averaged power rates and the ANOVA due to the inclusion of unrealistic conditions (i.e., the 100% careless item condition). The averaged power results may be somewhat biased upward by these unrealistic simulation conditions, which account for 48 out of the 192 total conditions. For example, it was found that excluding the 100% careless item conditions resulted in an approximately 22% decrease in statistical power compared to the scenario that included the 100% conditions. Nevertheless, to the best of our knowledge, there is currently no existing statistical approach specifically designed to detect careless respondents in MFC data. Although the I_z statistic in the TIRT model may not be flawless and yield varying power results under different conditions, it still represents a significant advance given the lack of an alternative statistical method available.

Practical Considerations

Practitioners and researchers must understand the potential indirect effects of item parameter estimation errors on the performance of the I_z person-fit. Organizations or test developers typically use pre-calibrated item parameters based on large, clean, and normative population data, and they score the test-takers' response data by fixing the item parameters. This method has been advocated for years in numerous testing applications (McDonald, 1982; Sass et al., 2008). We recommend that test developers use their pre-calibrated TIRT item parameters (obtained from MFC data) when computing person-fit statistics to maximize the performance of I_z person-fit.

However, this recommendation may not always be feasible in research settings, as it frequently necessitates large sample sizes and multiple data collection efforts. If obtaining pre-calibrated item parameters is not feasible in a research setting, we suggest using the proposed I_z person-fit statistics in conjunction with alternative detection methods such as response time, bogus items, or instructed-response items. For instance, researchers can initially identify potentially careless respondents based on prior detection methods, and then exclude them from the original dataset. The TIRT item parameters can then be estimated using the data, and the resulting item parameters may be regarded as proxies for pre-calibrated item parameters. Researchers can then identify careless respondents by applying the I_z person-fit statistic to the original response data based on the pre-calibrated item parameters. Given that test development or empirical studies in research settings frequently involve multiple data collections, the I_z person-fit statistic based on the pre-calibrated item parameters can be used (as a proxy) to ensure the quality of future response data.

Although this research focused on proposing a new approach to detecting careless respondents in MFC data, it is important to note that we do not claim that this is the only effective method, but it is one of the effective methods available. Recently, Arthur et al. (2021) emphasized that “if the decision is to remove careless responders from the dataset, then researchers should not rely on only one technique to screen careless responders for removal” (p. 115). Researchers may consider other possible detection approaches when looking to detect careless respondents in MFC data. One practice would be the use of instructed response items (e.g., “Please rank the responses as 3rd, 1st, and 2nd for this item” or “Please rank all the responses as 3rd rank for this item”). The instructed response items are easy to implement and quite effective (Kam & Chan, 2018). Another possible practice would be the use of response time to identify careless respondents in MFC data. In this case, researchers should be cautious when determining an appropriate threshold, as the comparative decision-making process in MFC responses

differs from the absolute decision-making process in Likert-type measures.

Researchers may also consider a multiple-hurdle approach when implementing different indices to identify careless respondents in MFC data. In this approach, a cut score is selected for each hurdle, and respondents who fail to pass a hurdle are eliminated (Arthur et al., 2021; Curran, 2016). Given that the robustness of each index's misclassification is a key consideration when using the multiple-hurdle approach (Arthur et al., 2021; Goldammer et al., 2020), the I_z person-fit statistic may be a suitable choice for an early hurdle in this process. As a result, researchers can improve their ability to accurately identify careless respondents while minimizing the risk of misclassification errors by combining multiple indices.

Limitations and Future Research

There are several limitations that need to be addressed. First, our results suggest that participants exhibited similar or higher levels of careless responding when responding to the MFC measure compared to the Likert measure. However, it is important to note that both measures yielded relatively low scores for careless responding behavior, ranging from “rarely” to “sometimes”, which may not be of practical significance. These relatively low scores may be due to their socially desirable responses. Acknowledging careless responding behavior could evoke negative feelings or guilt, especially since they were given credit for their participation. Following the social exchange theory, participants may have been motivated to provide more socially desirable responses to fulfill their perceived responsibility and avoid being perceived as uncooperative. Future research may benefit from incorporating more objective measures, such as response time, to assess careless response behavior rather than relying solely on self-reported subjective measures. Nevertheless, the MFC measures exhibited greater variability and higher frequency of careless responding than the Likert measures. This discrepancy may be attributed to the studied construct, an undesirable dark personality. From the test-taker's perspective, it may be more difficult to select the most undesirable option from a group of equally undesirable statements than to simply determine the degree of agreement from undesirable single-statement Likert items. Future research should examine the various factors that contribute to the different distributions of careless responding behaviors observed between Likert and MFC measures.

Second, this study used a dark personality MFC measure rather than a bright personality MFC measure. The rationale for this selection was based on the unique characteristics of the dark personality MFC measure, which are associated with undesirable constructs and the forced-choice format, potentially rendering them more effective in capturing

careless responses. However, it remains uncertain whether the findings of our study can be generalized to bright personality traits or other psychological constructs. Therefore, future research should examine how different psychological constructs may differentially influence the occurrence of careless responses on MFC measures, as well as explore the underlying mechanisms that drive these effects.

Third, we used a single-item measure to assess careless responding behavior. While the use of single-item measures is not uncommon in research investigating careless responding, as pointed out by an anonymous reviewer, concerns about the validity of such measures may arise. It is important to consider whether a single item adequately captures the careless responding behavior. To address these concerns, future research should consider providing a clear definition of careless responding and including relevant examples in the question itself. This approach would help ensure that a single-item measure captures the target construct, thereby improving the effectiveness of the single-item measure. In addition, researchers can also consider using multi-item measures or alternative measurement strategies to improve the measurement of careless responding behavior. By employing more comprehensive scales or alternative approaches, a more thorough and accurate assessment of careless responding behavior can be achieved.

Finally, while our simulations explored 192 experimental conditions, this may still be considered a limited representation of the many possibilities that may arise in real MFC testing situations. For instance, to simulate careless responses, we randomly generated MFC responses of the TIRT model, but other forms of careless responding may occur in actual MFC testing applications (e.g., same rankings appear consecutively: 1–2–3, 1–2–3, 1–2–3, etc.). In addition, we limited our investigation to the TIRT model. However, various IRT models (e.g., MUPP [Stark et al., 2005], GGUM-RANK [Lee et al., 2019a, 2019b], ZG-MUPP [Joo et al., 2023]) have been developed and are available for modeling MFC measures in applied and research settings, thus future research should investigate the careless response detection methods for various IRT models. In sum, it would be beneficial for future research to broaden the scope of the simulations to represent the diverse conditions that occur in actual testing.

Conclusion

In conclusion, the detection of careless respondents holds significant potential for improving the psychometric properties of response data and the quality of organizational decision-making. However, the issue of careless responses

in MFC data has remained unexplored. As the first study that explicitly investigates this matter, our research fills a critical gap in the literature. By equipping researchers and practitioners with the analytical tools and knowledge to effectively address careless responses in their studies, we aim to maximize the quality of MFC data and increase the validity and reliability of MFC data, ultimately improving the quality of organizational decision-making.

Appendix: A tutorial of l_z person-fit of the TIRT model

A tutorial is provided to assist researchers and practitioners in implementing the l_z person-fit statistic of the TIRT model for pair and triplet MFC data. This tutorial employs the R and Mplus programs. Although only a triplet MFC example is provided in this tutorial, the same procedures can also be applied to pair MFC measures. Readers can find the R source codes and relevant files in the online [supplementary materials](#). The example utilized a simulated response dataset with 1000 observations and five dimensions.

In step 1, users are recommended to create a working directory and place all pertinent files within it. This will ensure that all materials required for analysis are easily accessible and organized in one location.

```
setwd("C:/directory")
```

In step 2, install and load the “MplusAutomation” package. This package provides various useful functions for automating the analysis process with Mplus and is necessary for running the l_z person-fit statistic in R.

```
install.packages("MplusAutomation")
```

```
library(MplusAutomation)
```

In step 3, import the source code for calculating l_z person-fit in either the “pair” or “triplet” format. We note that “tirt.lz.pair.R” and “tirt.lz.triplet.R” are R source codes for using the l_z person-fit statistic with the “pair” and “triplet” format of MFC data. Both options are available, but we will only be presenting an example using the “triplet” format in this tutorial.

```
source("tirt.lz.triplet.R")
```

In step 4, specify information about the dataset, including the number of statements, binary responses, blocks, dimensions, and sample size. This information

is necessary for the analysis process and will be used to specify the parameters of the model.

```
n.statements <- 60    # Number of statement

n.responses <- 60      # Number of binary responses of TIRT model

n.blocks <- 20         # Number of item blocks

n.samples <- 876       # Sample size

n.dimensions <- 5      # Number of dimensions
```

In step 5, specify the file names for the input and output files that will be used in the analysis process. They include i) the file containing statement dimension information ("StateDim3_5D.txt"), ii) the file containing the output

from fitting the TIRT model in Mplus ("mptext1.out"), iii) the file containing the TIRT scores generated by Mplus ("Scores_5D.txt").

```
dimfile <- "StateDim3_5D.txt"    # Statement dimension file name

outfile <- "mptext1.out"         # TIRT Mplus output file name

scorefile <- "Scores_5D.txt"     # TIRT score file name
```

In step 6, execute the l_z function to calculate the l_z person-fit statistic for your data. This function takes the various parameters and files that users have specified in previous steps and produces a vector of l_z values for each respondent.

```
lz <- tirt.lz.triplet(n.statements, n.responses, n.blocks,
                     n.samples, n.dimensions, dimfile,
                     outfile, scorefile)
```

Here, we present the l_z values for only the first 5 respondents.

```
lz
[1] 0.778829862
[2] 0.252360107
[3] 0.673301523
[4] 0.583847476
[5] 0.006812512
```

In step 7, determine the empirical cutoff value. To compute the empirical critical value, "normal" binary response data of the TIRT model needs to be first generated based on the pre-calibrated item parameters (e.g., item loadings, thresholds, and factor correlations) in shown mptext1.out. We note that users can find Mplus example files "mptext3.out" and "mptext4.out" in the [supplementary materials](#) for the data generation of the TIRT model. In the provided files ("mptext3.out" and "mptext4.out"). The multivariate normal distribution was used to generate 1000 theta values ("mptext3.out"). Using the theta values created by "mptext3.out" and the pre-calibrated item parameters (i.e., factor loadings, thresholds, factor correlations) in "mptext1.out", binary response data for the TIRT model were then generated. Through this procedure, 20-triplet binary response data was generated. The estimated TIRT item parameters ("mptext2.out") and latent trait scores ("Scores_5D_empirical.txt") of the generated data were used to calculate the l_z values for each simulee. The l_z values derived from the generated data were sorted in ascending order, and the lower one-tailed value corresponding

to the 5th percentile of the I_z distribution was identified and used as the empirical I_z critical value. Examples are shown below:

```
# Specify TIRT output file obtained from the generated data
outfile.empirical <- "mptext2.out"

# Specify TIRT score file obtained from the generated data
scorefile.empirical <- "Scores_5D_empirical.txt"

#Compute empirical lz values and save as an emp.lz

emp.lz <- tirt.lz.triplet(n.statements, n.responses, n.blocks,
                        n.samples, n.dimensions, dimfile,
                        outfile.empirical, scorefile.empirical)

# Find the cutoff value

emp.cutoff <- quantile(emp.lz,0.05)

emp.cutoff
```

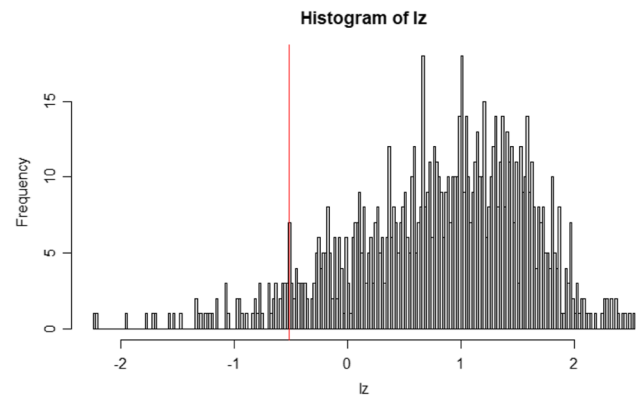
```
> emp.cutoff
      5%
-0.5124796
```

Step 8 examines the distribution of the I_z person-fit values with the empirical critical value. This will allow users to visually evaluate the distribution and the misfit cases. We note that the red vertical line indicates the empirical critical value.

```
#Create a histogram of lz values

hist(lz, breaks=200)

abline(v=emp.cutoff, col ="red", lty =1)
```



In step 9, identify which respondents are considered person-misfit cases. To do this, users will compare the I_z person-fit statistic for each respondent to the empirical critical value determined in Step 7. Any respondents whose I_z person-fit statistic is less than the empirical critical value can be considered careless respondents.

```
case_id <- which(lz < emp.cutoff)

lz_value <- lz[case_id]

data.frame(case_id,lz_value)
```

Here, we present only the first 5 detected respondents.

```
> data.frame(case_id,lz_value)
  case_id  lz_value
1      11 -0.5329325
2      33 -0.9726769
3      36 -1.1438346
4      49 -0.5923739
5      68 -0.5412359
```

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10869-023-09911-w>.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

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